

evolved with the Internet. ISOC promotes research and activities. IAB is the technical advisor to the ISOC. IETF is a forum of working groups responsible for operational problems. IRTF is a forum of working groups focusing on long-term research topics.

1.7 PRACTICE SET

1.7.1 Quizzes

A set of interactive quizzes for this chapter can be found on the book website. It is strongly recommended that the student take the quizzes to check his/her understanding of the materials before continuing with the practice set.

1.7.2 Questions

- Q1-1.** Identify the five components of a data communications system.
- Q1-2.** What are the three criteria necessary for an effective and efficient network?
- Q1-3.** What are the advantages of a multipoint connection over a point-to-point one?
- Q1-4.** What are the two types of line configuration?
- Q1-5.** Categorize the four basic topologies in terms of line configuration.
- Q1-6.** What is the difference between half-duplex and full-duplex transmission modes?
- Q1-7.** Name the four basic network topologies, and cite an advantage of each type.
- Q1-8.** For n devices in a network, what is the number of cable links required for a mesh, ring, bus, and star topology?
- Q1-9.** What are some of the factors that determine whether a communication system is a LAN or WAN?
- Q1-10.** What is an internet? What is the Internet?
- Q1-11.** Why are protocols needed?
- Q1-12.** In a LAN with a link-layer switch (Figure 1.8b), Host 1 wants to send a message to Host 3. Since communication is through the link-layer switch, does the switch need to have an address? Explain.
- Q1-13.** How many point-to-point WANs are needed to connect n LANs if each LAN should be able to directly communicate with any other LAN?
- Q1-14.** When we use local telephones to talk to a friend, are we using a circuit-switched network or a packet-switched network?
- Q1-15.** When a resident uses a dial-up or DLS service to connect to the Internet, what is the role of the telephone company?
- Q1-16.** What is the first principle we discussed in this chapter for protocol layering that needs to be followed to make the communication bidirectional?
- Q1-17.** Explain the difference between an Internet draft and a proposed standard.
- Q1-18.** Explain the difference between a required RFC and a recommended RFC.
- Q1-19.** Explain the difference between the duties of the IETF and IRTF.

1.7.3 Problems

- P1-1.** What is the maximum number of characters or symbols that can be represented by Unicode?
- P1-2.** A color image uses 16 bits to represent a pixel. What is the maximum number of different colors that can be represented?
- P1-3.** Assume six devices are arranged in a mesh topology. How many cables are needed? How many ports are needed for each device?
- P1-4.** For each of the following four networks, discuss the consequences if a connection fails.
 - a.** Five devices arranged in a mesh topology
 - b.** Five devices arranged in a star topology (not counting the hub)
 - c.** Five devices arranged in a bus topology
 - d.** Five devices arranged in a ring topology
- P1-5.** We have two computers connected by an Ethernet hub at home. Is this a LAN or a WAN? Explain the reason.
- P1-6.** In the ring topology in Figure 1.7, what happens if one of the stations is unplugged?
- P1-7.** In the bus topology in Figure 1.6, what happens if one of the stations is unplugged?
- P1-8.** Performance is inversely related to delay. When we use the Internet, which of the following applications are more sensitive to delay?
 - a.** Sending an e-mail
 - b.** Copying a file
 - c.** Surfing the Internet
- P1-9.** When a party makes a local telephone call to another party, is this a point-to-point or multipoint connection? Explain the answer.
- P1-10.** Compare the telephone network and the Internet. What are the similarities? What are the differences?

1.8 SIMULATION EXPERIMENTS

1.8.1 Applets

One of the ways to show the network protocols in action or visually see the solution to some examples is through the use of interactive animation. We have created some Java applets to show some of the main concepts discussed in this chapter. It is strongly recommended that the students activate these applets on the book website and carefully examine the protocols in action. However, note that applets have been created only for some chapters, not all (see the book website).

1.8.2 Lab Assignments

Experiments with networks and network equipment can be done using at least two methods. In the first method, we can create an isolated networking laboratory and use

Another model that defines protocol layering is the Open Systems Interconnection (OSI) model. Two layers in the OSI model, session and presentation, are missing from the TCP/IP protocol suite. These two layers were not added to the TCP/IP protocol suite after the publication of the OSI model. The application layer in the suite is usually considered to be the combination of three layers in the OSI model. The OSI model did not replace the TCP/IP protocol suite because it was completed when TCP/IP was fully in place and because some layers in the OSI model were never fully defined.

2.5 PRACTICE SET

2.5.1 Quizzes

A set of interactive quizzes for this chapter can be found on the book website. It is strongly recommended that the student take the quizzes to check his/her understanding of the materials before continuing with the practice set.

2.5.2 Questions

- Q2-1.** What is the first principle we discussed in this chapter for protocol layering that needs to be followed to make the communication bidirectional?
- Q2-2.** Which layers of the TCP/IP protocol suite are involved in a link-layer switch?
- Q2-3.** A router connects three links (networks). How many of each of the following layers can the router be involved with?
a. physical layer **b.** data-link layer **c.** network layer
- Q2-4.** In the TCP/IP protocol suite, what are the identical objects at the sender and the receiver sites when we think about the logical connection at the application layer?
- Q2-5.** A host communicates with another host using the TCP/IP protocol suite. What is the unit of data sent or received at each of the following layers?
a. application layer **b.** network layer **c.** data-link layer
- Q2-6.** Which of the following data units is encapsulated in a frame?
a. a user datagram **b.** a datagram **c.** a segment
- Q2-7.** Which of the following data units is decapsulated from a user datagram?
a. a datagram **b.** a segment **c.** a message
- Q2-8.** Which of the following data units has an application-layer message plus the header from layer 4?
a. a frame **b.** a user datagram **c.** a bit
- Q2-9.** List some application-layer protocols mentioned in this chapter.
- Q2-10.** If a port number is 16 bits (2 bytes), what is the minimum header size at the transport layer of the TCP/IP protocol suite?
- Q2-11.** What are the types of addresses (identifiers) used in each of the following layers?
a. application layer **b.** network layer **c.** data-link layer

- Q2-12.** When we say that the transport layer multiplexes and demultiplexes application-layer messages, do we mean that a transport-layer protocol can combine several messages from the application layer in one packet? Explain.
- Q2-13.** Can you explain why we did not mention multiplexing/demultiplexing services for the application layer?
- Q2-14.** Assume we want to connect two isolated hosts together to let each host communicate with the other. Do we need a link-layer switch between the two? Explain.
- Q2-15.** If there is a single path between the source host and the destination host, do we need a router between the two hosts?

2.5.3 Problems

- P2-1.** Answer the following questions about Figure 2.2 when the communication is from Maria to Ann:
- a.** What is the service provided by layer 1 to layer 2 at Maria's site?
 - b.** What is the service provided by layer 1 to layer 2 at Ann's site?
- P2-2.** Answer the following questions about Figure 2.2 when the communication is from Maria to Ann:
- a.** What is the service provided by layer 2 to layer 3 at Maria's site?
 - b.** What is the service provided by layer 2 to layer 3 at Ann's site?
- P2-3.** Assume that the number of hosts connected to the Internet at year 2010 is five hundred million. If the number of hosts increases only 20 percent per year, what is the number of hosts in year 2020?
- P2-4.** Assume a system uses five protocol layers. If the application program creates a message of 100 bytes and each layer (including the fifth and the first) adds a header of 10 bytes to the data unit, what is the efficiency (the ratio of application-layer bytes to the number of bytes transmitted) of the system?
- P2-5.** Assume we have created a packet-switched internet. Using the TCP/IP protocol suite, we need to transfer a huge file. What are the advantage and disadvantage of sending large packets?
- P2-6.** Match the following to one or more layers of the TCP/IP protocol suite:
- a.** route determination
 - b.** connection to transmission media
 - c.** providing services for the end user
- P2-7.** Match the following to one or more layers of the TCP/IP protocol suite:
- a.** creating user datagrams
 - b.** responsibility for handling frames between adjacent nodes
 - c.** transforming bits to electromagnetic signals
- P2-8.** In Figure 2.10, when the IP protocol decapsulates the transport-layer packet, how does it know to which upper-layer protocol (UDP or TCP) the packet should be delivered?
- P2-9.** Assume a private internet uses three different protocols at the data-link layer (L1, L2, and L3). Redraw Figure 2.10 with this assumption. Can we say that,

in the data-link layer, we have demultiplexing at the source node and multiplexing at the destination node?

- P2-10.** Assume that a private internet requires that the messages at the application layer be encrypted and decrypted for security purposes. If we need to add some information about the encryption/decryption process (such as the algorithms used in the process), does it mean that we are adding one layer to the TCP/IP protocol suite? Redraw the TCP/IP layers (Figure 2.4 part b) if you think so.
- P2-11.** Protocol layering can be found in many aspects of our lives such as air traveling. Imagine you make a round-trip to spend some time on vacation at a resort. You need to go through some processes at your city airport before flying. You also need to go through some processes when you arrive at the resort airport. Show the protocol layering for the round trip using some layers such as baggage checking/claiming, boarding/unboarding, takeoff/landing.
- P2-12.** The presentation of data is becoming more and more important in today's Internet. Some people argue that the TCP/IP protocol suite needs to add a new layer to take care of the presentation of data. If this new layer is added in the future, where should its position be in the suite? Redraw Figure 2.4 to include this layer.
- P2-13.** In an internet, we change the LAN technology to a new one. Which layers in the TCP/IP protocol suite need to be changed?
- P2-14.** Assume that an application-layer protocol is written to use the services of UDP. Can the application-layer protocol use the services of TCP without change?
- P2-15.** Using the internet in Figure 1.11 (Chapter 1) in the text, show the layers of the TCP/IP protocol suite and the flow of data when two hosts, one on the west coast and the other on the east coast, exchange messages.